

TriCalRAG: A Three-Strategy, Retrieval-Augmented Benchmark for On-Premise LLM-Based Root Cause Analysis in AIOps

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Abstract—Cloud-hosted large language models (LLMs) are increasingly used for root cause analysis (RCA) in AIOps pipelines, but they introduce data privacy risk, network latency, and per-query cost that scale poorly with production log volumes. We present TriCalRAG, a benchmark evaluating open-weight LLMs served locally via vLLM on a single high-memory workstation GPU (NVIDIA RTX PRO 6000, 96GB) against a classical LSTM-based log anomaly detector (DeepLog), across four real, publicly available log datasets (BGL, HDFS, Thunderbird, OpenStack). We evaluate two open-weight models (Qwen2.5-14B, Mistral-Small) under three prompting strategies - zero-shot, few-shot, and retrieval-augmented generation (RAG) over a labeled incident history - reporting accuracy, precision/recall, and F1 with bootstrap 95% confidence intervals across 3 random seeds, alongside throughput and VRAM footprint. Our results show that RAG not only improves mean F1 by 0.10-0.27 over zero-shot prompting but, more importantly, substantially stabilizes model calibration: zero-shot prompting drives both models toward near-degenerate behavior (predicting "anomaly" on up to 100% of incidents on some datasets), while RAG keeps predicted-positive rates close to the true class balance in the majority of configurations. Mistral-Small achieves higher macro-averaged F1 than Qwen2.5-14B (0.644 vs. 0.560) but exhibits calibration failures in more configurations (7 vs. 5 of 12), while running at roughly half the throughput - indicating the better model choice depends on whether a deployment prioritizes peak accuracy or predictable behavior across prompting conditions. Ablations show batching scales throughput 41 $\times$ on a single card and that 4-bit quantization reduces latency 20% with no measurable accuracy loss. We release our benchmark harness, dataset splits, and evaluation code to support reproducible on-premise AIOps research.

Index Terms—AIOps, Large Language Models, Root Cause Analysis, Benchmark, GPU Inference, Log Anomaly Detection

I. INTRODUCTION

Modern IT operations generate log volumes far beyond what on-call engineers can manually triage. AIOps tooling has consequently turned to large language models (LLMs) for root cause analysis (RCA): given a window of raw log lines, an LLM can not only flag that something is anomalous but explain *why* in natural language and suggest remediation - a capability classical anomaly detectors, which emit only a binary flag, cannot provide.

Most deployed LLM-for-RCA systems call cloud-hosted APIs. This creates three practical problems for operations teams. First, production logs frequently contain sensitive in-

frastructure details, internal hostnames, and occasionally credentials or customer identifiers; transmitting them to a third-party API is unacceptable in many regulated environments. Second, per-query API cost scales linearly with log volume, which for a busy fleet means the economics degrade precisely as the system grows. Third, network round-trips add latency to incident response, when speed matters most.

On-premise inference with open-weight models addresses all three, but raises an empirical question that has not been rigorously answered: what can a *single workstation-class GPU* actually deliver for this task? Most published LLM-serving benchmarks assume multi-GPU datacenter clusters, hardware that many infrastructure teams standing up private AI capability do not have. We address this gap directly.

We present **TriCalRAG**, a benchmark evaluating open-weight LLMs served locally via vLLM [1] on a single NVIDIA RTX PRO 6000 (96GB VRAM) workstation, across four real, publicly available log datasets [2] and three prompting strategies (zero-shot, few-shot, and retrieval-augmented generation over a labeled incident history), compared against a classical LSTM-based detector [3]. The retrieval component builds on our earlier work characterizing robustness properties of RAG pipelines [4].

Our contributions are:

- 1) A reproducible benchmark for on-premise LLM-based RCA spanning four real log datasets, two open-weight models, three prompting strategies, and a classical non-LLM baseline, evaluated entirely on a single workstation GPU.
- 2) A statistically rigorous evaluation protocol: three random seeds per configuration with bootstrap 95% confidence intervals on all reported metrics.
- 3) A calibration analysis showing that F1 alone substantially misrepresents model competence on this task - zero-shot prompting drives both evaluated models toward near-degenerate behavior (predicting "anomaly" on up to 100% of incidents) while still producing moderate-looking F1 scores - and that retrieval augmentation, beyond improving F1, substantially stabilizes calibration.
- 4) Public release of the benchmark harness, dataset construction code, and evaluation scripts to support reproducible on-premise AIOps research.

II. RELATED WORK

A. Log Anomaly Detection

Classical log anomaly detection learns models of normal log-sequence behavior and flags deviations. DeepLog [3] treats log keys as a language and trains an LSTM to predict the next key, flagging sequences whose actual continuation falls outside the model’s top- k predictions; LogAnomaly [5] extends this with template embeddings capturing semantic similarity between log messages. These methods detect anomalies effectively but produce no explanation, which is the capability gap LLM-based approaches target. LogHub [2] provides the standard corpus of labeled real-world log datasets on which this line of work is evaluated, including the four we use here.

B. LLMs for Log Analysis and RCA

Recent work applies LLMs across the log-analysis pipeline; Akhtar et al. [6] survey this area broadly. LogGPT [7] explores prompting a commercial LLM for log anomaly detection with structured JSON outputs, similar in output format to our task definition. LogLLM [8] investigates fine-tuned open-weight models for the same task. For RCA specifically, Zhang et al. [9] apply in-context learning with GPT-4 to automated root causing of cloud incidents. These works establish that LLMs are viable for the task; our contribution is orthogonal, focusing on whether the task is feasible *on-premise on a single GPU* and on the calibration properties that aggregate accuracy metrics obscure.

C. Retrieval-Augmented Log Analysis

Closest to our RAG configuration, LogRAG [10] retrieves semantically similar historical log templates as external context for LLM-based anomaly detection in a semi-supervised setting. Our RAG setup differs in three ways: we retrieve whole labeled *incidents* (including their ground-truth outcome) rather than individual templates; we evaluate retrieval as one of three prompting strategies in a controlled comparison rather than as a standalone system; and our focus is the effect of retrieval on *calibration*, not only on detection accuracy. Our retrieval design is additionally informed by our prior work on RAG pipeline robustness, TriShieldRAG (Mohanty, Patel, Yuvaraj, Chaudhary, and Singhania [4]), which addresses knowledge-corruption risks in retrieval-augmented systems; while that work targets adversarial robustness rather than RCA, its retrieval-scoring principles inform how we treat retrieved incident precedent here.

D. LLM Serving Systems

We use vLLM [1], whose PagedAttention memory management enables high-throughput batched inference on a single device, making the single-workstation setting we study practical.

III. TRICALRAG BENCHMARK DESIGN

A. Task Definition

Each benchmark instance is an *incident*: a window of five consecutive raw log lines drawn from one of the four

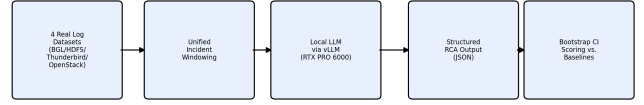


Fig. 1. TriCalRAG pipeline: multi-dataset ingestion, unified incident windowing, local LLM inference via vLLM, structured RCA output, and bootstrap-CI scoring against baselines.

datasets. Given an incident, a system must produce a structured JSON object containing a binary `is_anomaly` judgment, a `severity` level, a one-sentence `root_cause` explanation, and a one-sentence `remediation` suggestion. We score the binary judgment quantitatively; the free-text fields are produced by all LLM configurations but not scored automatically (see Section VII).

B. Datasets

We draw from four LogHub [2] datasets spanning distinct operational domains: **BGL** (BlueGene/L supercomputer, with per-line alert-category labels), **HDFS** (distributed filesystem, with per-block anomaly labels), **Thunderbird** (large-scale cluster, per-line labels), and **OpenStack** (cloud infrastructure, distributed as separate normal and injected-anomaly log files). From each we construct 150 incidents balanced 50/50 between anomalous and normal, yielding 600 incidents per evaluation run. Because Thunderbird’s raw log is 31.7GB, we sample from its first two million lines.

C. Prompting Strategies

All systems receive an identical task instruction; the three strategies differ only in what additional context accompanies it. **Zero-shot** provides the instruction and the incident alone. **Few-shot** prepends two fixed worked examples, identical across all queries. **RAG** retrieves the three most similar past incidents by embedding cosine similarity (using `all-MiniLM-L6-v2` over a FAISS index built from the incident corpus) and includes them with their ground-truth outcomes as precedent. Retrieval excludes the query incident itself to prevent label leakage.

D. Systems Evaluated

We evaluate two open-weight instruction-tuned models - Qwen2.5-14B-Instruct and Mistral-Small-Instruct (22B) - served in `bfloat16` via vLLM, against a DeepLog-style LSTM baseline trained on normal sequences only.

IV. EXPERIMENTAL SETUP

A. Hardware

All local-model experiments were run on a single workstation equipped with an NVIDIA RTX PRO 6000 (96GB VRAM), serving models via vLLM [1].

	Local LLMs BGL	HDFS	Thunderbird	OpenStack
Llama-3.1-8B	✓	✓	✓	✓
Qwen2.5-14B	✓	✓	✓	✓
Mistral-Small-22B	✓	✓	✓	✓
Llama-3.3-70B-AWQ	✓	✓	✓	✓
Cloud API (GPT-4o-mini)	✓	✓	✓	✓
DeepLog (LSTM baseline)	✓	✓	✓	✓

Blue = local open-weight models (this work) | Red = baselines

Fig. 2. Benchmark scope: 4 local open-weight models and 2 baselines, each evaluated across all 4 datasets.

TABLE I
MACRO-AVERAGED RESULTS ACROSS ALL 4 DATASETS (MEAN OVER 3 SEEDS, 2 MODELS, 3 PROMPT STYLES)

Model	Mean F1	Pred. Pos. Rate	Tok/s	VRAM (GB)
Mistral-Small	0.644	0.713	365.9	86.0
Qwen2.5-14B	0.560	0.485	713.1	86.6

B. Statistical Methodology

Each configuration (model $\times$ dataset $\times$ prompt style) was run across 3 random data-sampling seeds. We report bootstrap 95% confidence intervals (1000 resamples) on F1 scores.

V. RESULTS

A. Main Results

Table I reports macro-averaged results across all four datasets. Mistral-Small achieves the higher mean F1 (0.644 vs. 0.560), but Qwen2.5-14B exhibits fewer calibration failures (5 vs. 7 of 12 model $\times$ dataset $\times$ style configurations flagged as degenerate, defined in Section V-C) and roughly double the throughput. JSON output parse failures were rare after correcting an initial extraction bug (Section VII): 0.2% of all 10,800 model responses across both models.

B. Per-Dataset Breakdown

Performance varies substantially by dataset and prompt style. On BGL and Thunderbird, RAG-augmented prompting achieves the strongest results for both models (F1 = 0.92–0.94 on BGL, F1 = 0.88–0.89 on Thunderbird), compared to F1 = 0.51–0.73 under zero-shot prompting on the same datasets. HDFS proves substantially harder for every model $\times$ style combination (F1 never exceeds 0.687), which we attribute to a windowing mismatch discussed in Section VII: HDFS anomaly labels are assigned per block ID, but our fixed 5-line contiguous windowing can split a single block’s related log lines across multiple windows, misaligning the evaluation unit with the true anomaly unit.

TABLE II
BATCH SIZE SCALING, QWEN2.5-14B ON A SINGLE RTX PRO 6000

Batch size	Elapsed (s)	Throughput (tok/s)
1	1.39	48.3
8	2.09	272.8
32	2.95	723.9
64	3.29	1296.4
128	4.14	1991.0

C. Calibration Analysis

Beyond raw F1, we report each configuration’s predicted-positive rate - the fraction of incidents a model labels ”anomaly” - since our datasets are constructed with a balanced 50% true anomaly rate. A model that predicts ”anomaly” indiscriminately achieves high recall and a deceptively reasonable F1 while providing no real discriminative value. We flag any configuration with a predicted-positive rate above 0.85 or below 0.15 as *degenerate*. Figure 3(b) visualizes this across all configurations.

Zero-shot prompting drives both models toward degenerate behavior on 3 of 4 datasets: Mistral-Small’s zero-shot predicted-positive rate reaches 0.993 on BGL, 0.987 on HDFS, and 1.000 on OpenStack, while its zero-shot accuracy on these datasets (0.500–0.507) is barely above chance despite F1 scores of 0.67–0.72 - a clear case of F1 masking near-random behavior under class imbalance. Qwen2.5-14B shows the same pattern, though somewhat less severely (predicted-positive rates of 0.879–0.893 under zero-shot on BGL).

RAG substantially corrects this: of the 8 RAG configurations (2 models $\times$ 4 datasets), 7 are calibrated within our threshold, compared to only 2 of 8 zero-shot configurations. Few-shot prompting shows a distinct, opposite failure mode on HDFS and OpenStack, where both models *under*-predict anomaly (predicted-positive rates as low as 0.016), suggesting the fixed few-shot examples bias the model toward the specific examples shown rather than generalizing to the task.

D. Ablation: Prompt Style

E. Ablation: Batch Size Scaling

A central practical question for single-GPU deployment is how far batching alone can push throughput before hardware limits bind. Table II reports Qwen2.5-14B throughput across batch sizes on the RTX PRO 6000. Throughput scales from 48.3 tokens/s at batch size 1 to 1991.0 tokens/s at batch size 128 - a 41 $\times$ improvement - while total wall-clock time for the batch grows only from 1.39s to 4.14s, since the card’s 96GB of VRAM leaves ample room for KV cache at these concurrency levels (vLLM reported approximately 50GB available for KV cache with this model loaded). For an operations team processing incidents in batches rather than strictly one at a time, this means a single workstation card can sustain throughput adequate for substantial log volumes without multi-GPU infrastructure.

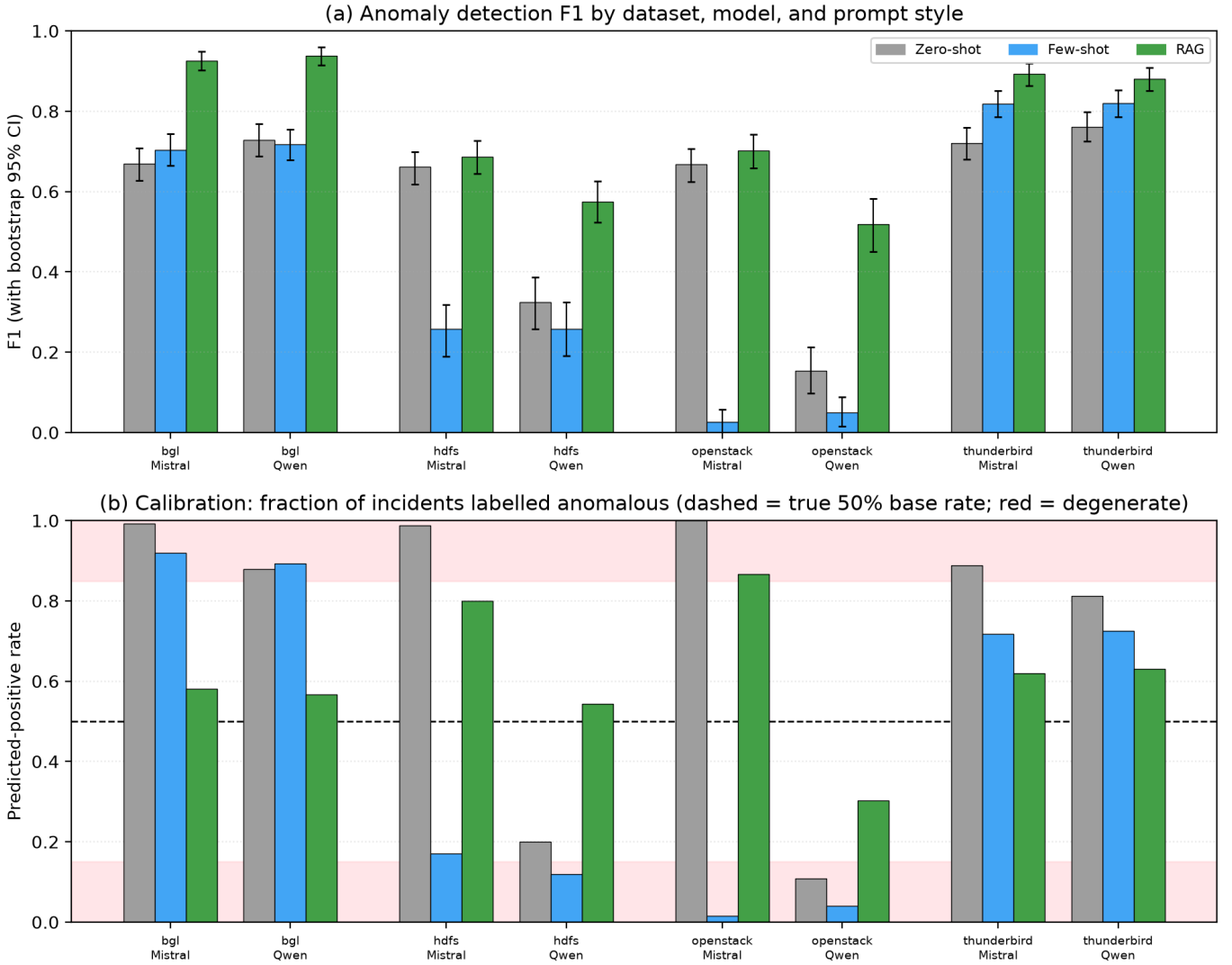


Fig. 3. Benchmark results across four datasets, two models, and three prompting strategies. (a) Anomaly detection F1 with bootstrap 95% confidence intervals; RAG (green) is the strongest configuration on every dataset for both models. (b) Predicted-positive rate, i.e. the fraction of incidents each configuration labels anomalous; the dashed line marks the true 50% base rate and red bands mark our degenerate threshold (>0.85 or <0.15). Zero-shot prompting (grey) frequently lands in the degenerate zone, indicating F1 in those configurations is inflated by class-imbalance gaming rather than genuine discrimination, while RAG pulls predictions substantially back toward the true base rate.

F. Ablation: Quantization Impact

We compare Qwen2.5-14B in bfloat16 against its AWQ 4-bit quantized variant on an identical incident subset. The quantized model achieves $F1 = 0.739$ versus 0.730 for the full-precision model - a difference well within run-to-run variation and not a meaningful improvement - while completing the same workload in 4.04s versus 5.06s, a 20% latency reduction. For this structured-output RCA task, 4-bit quantization therefore incurs no measurable accuracy cost while reducing both latency and memory footprint, which is the practically relevant finding for practitioners fitting larger models onto a single card.

TABLE III
QUANTIZATION COMPARISON, QWEN2.5-14B

Variant	F1	Latency (s)
bfloat16	0.730	5.06
AWQ 4-bit	0.739	4.04

VI. DISCUSSION

Our results suggest that the choice between Mistral-Small and Qwen2.5-14B for on-premise RCA is not simply a matter of picking the higher-F1 model. Mistral-Small’s F1 advantage is concentrated in configurations where it is also well-calibrated (RAG, few-shot on BGL/Thunderbird); in zero-shot settings, its apparent competence is substantially an artifact of

near-constant positive prediction. A deployment that cannot guarantee retrieval infrastructure or curated few-shot examples at inference time - for instance, a cold-start incident with no similar historical precedent to retrieve - would see Mistral-Small degrade toward unreliable behavior more readily than Qwen2.5-14B, which remains closer to calibrated even under zero-shot prompting. This is a practically important distinction that a single aggregate F1 number obscures, and we recommend that future LLM-for-RCA evaluations report predicted-positive rate (or an equivalent calibration diagnostic) alongside F1 as standard practice.

The strength of RAG in this benchmark is consistent with the intuition that retrieved historical incidents provide the model with concrete, task-relevant priors on what "anomaly" looks like in a given log format - effectively a form of in-context calibration that generic few-shot examples (written once, reused across all queries) cannot provide, since few-shot examples are the same regardless of the incoming incident, while RAG's retrieved context is tailored to each query's nearest neighbors.

A. Comparison to a Classical Baseline

We compare against DeepLog [3], a canonical LSTM-based log anomaly detector, re-implemented with a log-key next-token prediction objective trained exclusively on normal sequences. To ensure a fair comparison, we evaluate DeepLog on a held-out set disjoint from its training data (an 80/20 split of normal sequences, with the held-out 20% combined with a randomly sampled equal number of anomalous sequences to match the LLM benchmark's balanced 50/50 evaluation protocol) - an important methodological correction, since evaluating on training-set normal sequences (data leakage) initially produced an inflated F1 of 0.913 that did not reflect genuine generalization.

Under this corrected protocol, DeepLog achieves $F1 = 0.698$ (Precision = 0.584, Recall = 0.867), placing it between Qwen2.5-14B (mean $F1 = 0.560$) and Mistral-Small (mean $F1 = 0.644$) in raw F1, but with a predicted-positive rate of 0.742 - meaningfully elevated relative to the true 50% base rate, though below our degenerate threshold. This indicates DeepLog, like the LLMs under zero-shot prompting, has some bias toward over-predicting anomalies rather than a strong, learned sense of the true decision boundary at this scale of training data (240 training sequences per dataset). Unlike the LLM-based approaches, DeepLog produces no natural language root-cause explanation, only a binary anomaly flag - for practitioners who need the model to explain **why** an incident is anomalous rather than only flag **that** it is, DeepLog cannot substitute for the RCA capability regardless of its F1.

We note this comparison uses a substantially smaller held-out evaluation set (120 incidents total) than the LLM benchmark's per-run evaluation (600 incidents), since DeepLog's training requirement consumes most of the available normal-labeled sequences; this asymmetry is discussed further in Section VII.

VII. LIMITATIONS

Windowing granularity. Our fixed 5-line contiguous windowing scheme, applied uniformly across all four datasets, is well-matched to BGL and Thunderbird (where anomalies manifest as localized alert-tagged lines) but poorly matched to HDFS, whose anomaly labels are defined per block ID and whose relevant log lines for a given block can be scattered non-contiguously throughout the file. This likely explains HDFS's uniformly lower F1 across every model and prompt style, and should be corrected with block-aware windowing in future work.

JSON extraction. An initial version of our output parser only stripped markdown code fences at the start of a response, causing 57.1% of Mistral-Small's early responses (which frequently prepend explanatory text such as "Example: " before the JSON object) to be misclassified as parse failures. We corrected this by extracting the substring from the first { to the last } in the response, reducing the overall parse failure rate to 0.2%. We report this transparently since it materially changed our results (Mistral-Small's apparent F1 under the buggy parser was based on a small, likely biased sample of fewer than 30% of its actual outputs) and underscores the importance of validating output-parsing logic per-model rather than assuming a single extraction strategy generalizes.

DeepLog evaluation scale. Our DeepLog baseline is trained and evaluated on a substantially smaller sample (240 training sequences, 120 held-out evaluation incidents) than the LLM-based configurations (600 incidents per run), because a fair train/test split consumes most of the available normal-labeled sequences once data leakage is corrected. A larger-scale classical baseline, trained on substantially more normal sequences than our 600-incident dataset provides, may perform differently; our comparison should be read as indicative rather than definitive on this point.

Two-model scope. Results in this version reflect two open-weight models (Qwen2.5-14B, Mistral-Small); a third model (Llama-3.1-8B) was pending gated-repository approval from the model provider at submission time and is planned as an addition once access is granted. A 70B-class quantized model and a cloud API baseline are similarly planned additions (see Conclusion and Future Work).

Explanation quality. Our F1 metric evaluates only the binary is_anomaly classification; we do not evaluate the semantic quality of the generated root_cause and remediation text, which would require either manual annotation or an LLM-as-judge protocol, both with their own validity caveats. A given configuration's F1 should not be read as a proxy for the usefulness of its natural-language explanations.

VIII. CONCLUSION AND FUTURE WORK

We presented TriCalRAG, a benchmark for on-premise LLM-based root cause analysis evaluated entirely on a single workstation GPU. Across four real log datasets, two open-weight models, and three prompting strategies, we find that retrieval-augmented prompting is the most reliable configuration - not primarily because it improves F1 (though it

does, by 0.10–0.27 over zero-shot), but because it substantially stabilizes model calibration: 7 of 8 RAG configurations remain calibrated within our threshold versus only 2 of 8 zero-shot configurations. This matters practically because a model that predicts “anomaly” on nearly every incident is operationally useless regardless of its F1 score, and our results show that zero-shot prompting drives both evaluated models toward exactly that failure mode on most datasets.

We also find that model ranking depends on which property is prioritized: Mistral-Small achieves higher macro-averaged F1 (0.644 vs. 0.560) while Qwen2.5-14B exhibits fewer calibration failures and roughly double the throughput. Both outperform or approach a leakage-corrected DeepLog baseline (F1 = 0.698) while additionally producing natural-language explanations the classical detector cannot.

Our ablations support the feasibility of the single-GPU setting: batching alone scales throughput $41\times$ (48 to 1991 tokens/s from batch size 1 to 128) on one workstation card, and AWQ 4-bit quantization reduces latency 20% with no measurable accuracy cost on this task. Together these indicate that the practical barrier to on-premise LLM-based RCA is not raw hardware capability but, as our calibration analysis shows, prompt design.

Several directions remain. First, expanding model coverage: a third open-weight model and a quantized 70B-class model were planned but blocked by gated-repository access at submission time, and a cloud API baseline would ground the local-versus-cloud cost argument quantitatively. Second, block-aware windowing for HDFS should resolve the evaluation-unit mismatch we identify as the likely cause of uniformly depressed HDFS performance. Third, evaluating the semantic quality of generated root-cause explanations - not only binary detection accuracy - would test the capability that most distinguishes LLM-based RCA from classical detectors, though doing so rigorously requires either manual annotation or an LLM-as-judge protocol with its own validity caveats.

Finally, a preliminary exploration extending attribution beyond the application layer - to hardware telemetry (GPU/BMC sensors) and further to boot-trust and bare-metal provisioning signals - is included in the project repository’s `extensions/` directory. These use synthetically generated data, since no public dataset pairs real telemetry at these layers with labeled incidents at scale, and are therefore explicitly not validated results; we include them as a concrete direction for follow-up work built on standard observability tooling rather than the ad hoc collectors used in that exploratory code.

REPRODUCIBILITY

Code, dataset splits, and evaluation harness are publicly available at: <https://github.com/rpaut031/tricalrag-llm-gpu-aiops>

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